Hallucination in Large Language Models: A Survey

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Abstract

This survey examines the problem of hallucination in large language models (LLMs), where hallucination is a models' tendency to generate fluent, plausible text that is factually incorrect or unsupported by any real sources. Hallucination has existed since early sequence-to-sequence architecture to now newer transformer-based LLMs, and I will review how the field has defined hallucination, how it has been evaluated, and the major classes of mitigation strategies that have emerged. We will go over intrinsic and extrinsic hallucination, faithfulness-oriented evaluation benchmarks, retrieval-augmented generation, training-based methods, and decoding-time interventions. Our goal is to provide a newcomer with a comprehensive picture of hallucinations in the field of NLP and Generative AI.

1 Introduction

Imagine yourself prompting an LLM like ChatGPT to help you with math homework. It gives you step-by-step explanations, and everything flows extremely well that you fully believe it… except that when you take a closer look, it is telling you that $1 + 1 = 7$. This one mistake cascades into you doing the entire problem wrong, and is just one of many ways that LLMs can hallucinate. But what is hallucination? Hallucination is the generation of text that is fluent, but factually incorrect or unsupported by any grounding source (Ji et al., 2023; Huang et al., 2023).[1] It is a long-standing issue that has existed before modern LLMs like ChatGPT, and continues to be stubborn in being present at varying degrees of use in LLMs. Hallucination is prevalent because if LLMs are to be used in high-stake domains, like medicine, law, or journalism, it not only becomes an accuracy issue, but a safety concern for users and their clients as well. Throughout this survey, I will dive deeper into hallucination, the historical context and where we stand today, how it can get evaluated, the multiple mitigation strategies, and the future challenges and direction of where it appears and how to combat it. The goal of this survey is to provide a newcomer with a comprehensive picture of hallucinations in the field of NLP and generative AI.

[1] Word count: **[2096]**

2 Defining Hallucination

This section will help set up definitions for the rest of the survey. Hallucination comes in multiple forms, most significantly being intrinsic and extrinsic hallucination. According to Ji et al. (2023), intrinsic hallucination is when the output of an LLM clearly contradicts the information explicitly given to the model. An example is when you ask Claude to summarize a long document you input, and it gives you a summary that contradicts its source document. Extrinsic hallucination is when a model's output adds information out of thin air; the additional info may or may not be true, but it is not immediately verifiable. These are immediate definitions, but there are overlying concepts that factor into hallucinations as well. These concepts are faithfulness and factuality. Faithfulness in this context portrays whether or not output stays consistent with provided context (Maynez et al., 2020). Does output align with context and prompts you have inputted—does it align with what you intended for the LLM to do? Factuality is whether or not output aligns with common world knowledge. Does the 25 year anniversary of a classic movie actually mean the movie was released in 2001, or 1999? Both faithfulness and factuality require different evaluation strategies, since a faithful summary can still be factually wrong in the real world.

Hallucination can mean different things depending on the context. In summarization, a model is given a source document and asked to condense it.

Hallucination here means the summary could add facts (extrinsic), or totally contradict the source it is given (intrinsic). In open-domain QA, the model just answers questions from memory. Hallucination here is saying the release year of the first Star Wars movie is 1999 instead of 1977. In a conversational chat, hallucination would be something like Google Gemini forgetting a fact the user just told it. In Neural Machine Translation (NMT), a model could translate an English sentence but hallucinate by outputting fluent Portuguese that is completely unrelated to the original sentence (detached), or by getting stuck repeating the same phrase in a loop (oscillatory) (Raunak et al., 2021). There is not an umbrella definition given all these different types of hallucinations, making cross-paper comparison difficult (Zhang et al., 2023).

3 Historical Context: From Seq2Seq to LLMs

Hallucination was not always just an "LLMs" problem. Before LLMs like ChatGPT became mainstream, there was the early NMT era from 2018–2021. During this time, seq2seq models hallucinate when inputs fall far from their training distribution. Raunak et al. (2021) connected NMT hallucinations under source perturbation to long-tail training data. Guerreiro et al. (2023b) expanded on this with detached hallucinations, where output ignores the source entirely, oscillatory hallucinations, where the model gets stuck repeating the same phrase in a loop, and others. This is particularly dangerous because readers who fully rely on a language translation and do not speak the original language cannot catch these mistakes.

These types of issues were reduced when Retrieval-Augmented Generation, or RAG, became popular around 2020. RAG retrieves relevant documents at inference time instead of relying purely on memorized knowledge, and is the first major architectural response to unreliable parametric knowledge (Lewis et al., 2020). It directly reduces hallucination in dialogue settings (Shuster et al., 2021).

Now and since 2022, LLMs have shown that hallucinations have not fully gone away, but have instead changed their appearance. This is due to field experts attempting to train models to be better, and this instead has made models better at appeasing people's preferences and hiding hallucinations. At OpenAI, there is a practice called RLHF done to produce outputs humans prefer (Ouyang et al., 2022). This is a double-edged sword because RLHF does make models more truthful, but it also teaches models to sound confident and helpful, which is why LLMs can give extremely confident hallucinations... like $1+1=7$. According to Kandpal et al. (2023), facts about obscure people, places, and events that seldom appear in training data can lead to models making things up, and scaling up those models does not fix it. Kadavath et al. (2022) notes that models actually have a good sense of when they are uncertain, but do not always communicate it. RLHF makes this worse by rewarding confident answers and hiding the fact that they might not know. Together, these points show that the LLM era did not solve hallucination, but instead changed its character, like a virus that mutates when attempted to be destroyed.

4 Evaluation: Benchmarks and Metrics

If hallucination is hard to define, measuring it is even harder. There are two main evaluations: faithfulness evaluation, which asks how closely an output sticks to a given source, and factuality evaluation, which asks whether the output is actually true in the world.

On the faithfulness side, Maynez et al. (2020) were among the first to systematically annotate how often summarization models introduce information that is not in their source documents. QAFactEval (Fabbri et al., 2022) built on this with a more scalable automated approach: generate questions from the source document, then check whether the model's output can answer them correctly. It is a significant step up from metrics like ROUGE, which measure word overlap and say nothing about factual accuracy. On the factuality side, TruthfulQA (Lin et al., 2022) contains 817 adversarially crafted questions that humans frequently get wrong due to common misconceptions, and the key finding is that larger models tend to be less truthful because they have absorbed more of the internet's bad information. HaluEval (Li et al., 2023a) built a large-scale dataset of human-annotated hallucination examples across QA, summarization, and dialogue, giving the field an actual training signal. FActScore (Min et al., 2023) goes the most granular: it decomposes a long-form response into individual atomic claims, retrieves Wikipedia evidence for each one, and scores them independently.

Some methods sidestep the need for external knowledge entirely. SelfCheckGPT (Manakul et al., 2023) works by sampling multiple outputs for the same prompt, where real facts stay consistent across samples while hallucinated ones vary. Guerreiro et al. (2023a) applied optimal transport to NMT hallucination detection without any labeled examples, staying competitive with supervised detectors. Gaur and Saunshi (2023) used symbolic math word problems as a proxy, catching cases where a model's stated reasoning does not match its final answer. On the vision-language side, POPE (Li et al., 2023b) converts hallucination evaluation into binary polling ("Is there a [object] in this image?"), which is more stable than parsing free-form captions.

The bottom line: no single metric captures everything. Faithfulness metrics miss world-knowledge errors, and factuality benchmarks miss context-grounding failures.

5 Mitigation Strategies

From these struggles emerge three distinct strategies, where each one diagnoses a different root cause and prescribes a different kind of fix.

5.1 Retrieval-Augmented Generation

The first strategy starts from a simple premise: if you cannot trust a model's parametric memory, give it something reliable to look at instead. Lewis et al. (2020) established the foundational RAG architecture where a retriever pulls relevant documents at query time instead of relying solely on memorized training data. As covered earlier, this produced significant factuality gains, and Shuster et al. (2021) confirmed that RAG models hallucinate measurably less than non-RAG ones in dialogue settings.

But RAG is not always the right call. Mallen et al. (2023) found that for popular, well-covered entities, a model's internal memory is reliable enough, and retrieval matters most for long-tail knowledge. Smart retrieval, knowing when to retrieve, outperforms always retrieving.

5.2 Training-Based Methods

The second strategy goes deeper: if hallucination comes from how a model was trained, fix the training. RLHF through Ouyang et al. (2022) does improve truthfulness, but can cause models to project confidence they do not have, which is exactly the failure mode described in the historical section.

Contrastive training offers a more targeted approach. van der Poel et al. (2022) showed that optimizing for mutual information between a source and its summary explicitly penalizes the model for adding unsupported content. HaluEval (Li et al., 2023a), beyond its role as a benchmark, also functions as a fine-tuning signal: supervised training on annotated hallucination examples can teach a model to recognize and avoid these patterns before it even generates an output. The main limitation is cost because human annotation at scale is slow and expensive, and domain-specific gains do not always transfer.

5.3 Decoding-Time Interventions

The third strategy argues that the model already has the knowledge it needs, and that the problem is in how you sample from it.

Contrastive decoding (Li et al., 2022) penalizes tokens that a weaker model assigns high probability to. The logic is that generic, highly predictable outputs are more likely to be hallucinated, and that pushing a good model away from those tokens produces more grounded outputs. DoLa (Chuang et al., 2024) applies a similar principle inside a single model where later transformer layers encode more factual knowledge than earlier ones, so amplifying that gap at inference time improves TruthfulQA scores by 12–17% with no fine-tuning or retrieval. Context-aware decoding (Shi et al., 2023) up-weights the influence of a retrieved context document at inference time, forcing the model to use it rather than defaulting to what it memorized. Waldendorf et al. (2024) extended contrastive decoding to multilingual NMT, showing the approach transfers beyond English QA. Azaria and Mitchell (2023) showed that a model's internal hidden states encode whether it is "lying," meaning white-box detection signals exist even without a reference source.

The advantage of decoding methods is that no retraining is needed. The limitation is added inference cost: they do not fix the root cause, only manage the output.

6 Open Challenges and Future Directions

After covering definitions, history, evaluation, and mitigation, a few big problems remain that the field has not cracked.

Evaluation is still the most foundational gap. No single metric handles both faithfulness and factuality at once, and FActScore is too expensive for most practitioners to run at scale. A growing trend of using LLMs as judges to evaluate other LLMs creates a cyclical problem. It is like two people who get their news from the same unreliable source trying to fact-check each other. Kadavath et al. (2022) showed that models are actually reasonably aware of their own uncertainty, but that honest uncertainty gets trained out by RLHF, which rewards confident-sounding answers.

On the knowledge side, Kandpal et al. (2023) showed that scaling up does not fix long-tail hallucination. A bigger model that barely saw a fact during training does not magically know it better, and in high-stakes domains like medicine, confidently guessing the wrong answer is not an option.

The scope of the problem is also expanding. POPE (Li et al., 2023b) and Himmi et al. (2024) show that hallucination in vision-language models introduces new failure modes that text-only evaluation cannot catch. Wang et al. (2024) proposed instruction contrastive decoding as a mitigation, but it is still early-stage work.

The core question remains: is hallucination a data problem, a training problem, or a decoding problem? The honest answer is probably all three, which is why there may never be a standalone solution.

7 Conclusion

Hallucination started as a small task-specific issue well before LLMs. It has since become one of the defining problems of modern NLP, with its own taxonomy, benchmarks, and mitigation strategies. The three strategies of mitigation—retrieval augmentation, training-based fixes, and decoding-time interventions—each address a different piece of the problem, and the best systems increasingly combine all three. But as both Zhang et al. (2023) and Huang et al. (2023) note in their surveys of the field, hallucination remains fundamentally unsolved. The field still cannot agree on a single definition or a unified way to measure it, and that gap itself acts as an obstacle to progress. The $1 + 1 = 7$ problem has not gone away. It has just gotten better at hiding.

References

Amos Azaria and Tom Mitchell. 2023. The internal state of an LLM knows when it's lying. *arXiv preprint arXiv:2304.13734*.

Yung-Sung Chuang, Yujia Xie, Hongyin Luo, Yoon Kim, James Glass, and Pengcheng He. 2024. DoLa: Decoding by contrasting layers improves factuality in large language models. In *The Twelfth International Conference on Learning Representations*.

Alexander R. Fabbri, Chien-Sheng Wu, Wenhao Liu, and Caiming Xiong. 2022. QAFACTEVAL: Improved QA-based factual consistency evaluation for summarization. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 2587–2601, Seattle, United States. Association for Computational Linguistics.

Vedant Gaur and Nikunj Saunshi. 2023. Reasoning in large language models through symbolic math word problems. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 5889–5903, Toronto, Canada. Association for Computational Linguistics.

Nuno M. Guerreiro, Pierre Colombo, Pablo Piantanida, and André Martins. 2023a. Optimal transport for unsupervised hallucination detection in neural machine translation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13766–13784, Toronto, Canada. Association for Computational Linguistics.

Nuno M. Guerreiro, Elena Voita, and André F. T. Martins. 2023b. Looking for a needle in a haystack: A comprehensive study of hallucinations in neural machine translation. *arXiv preprint arXiv:2208.05309*.

Anas Himmi, Guillaume Staerman, Marine Picot, Pierre Colombo, and Nuno M. Guerreiro. 2024. Enhanced hallucination detection in neural machine translation through simple detector aggregation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 18573–18583, Miami, Florida, USA. Association for Computational Linguistics.

Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. 2023. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *arXiv preprint arXiv:2311.05232*.

Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Shady El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Palermo, Noam Lad, Dustin Landau, and 10 others. 2022. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*.

Nikhil Kandpal, Haikang Deng, Adam Roberts, Eric Wallace, and Colin Raffel. 2023. Large language models struggle to learn long-tail knowledge. In *Proceedings of the 40th International Conference on Machine Learning*, pages 15696–15707.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.

Junyi Li, Xiaoxue Cheng, Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. 2023a. HaluEval: A large-scale hallucination evaluation benchmark for large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6449–6464, Singapore. Association for Computational Linguistics.

Xiang Lisa Li, Ari Holtzman, Daniel Fried, Percy Liang, Jason Eisner, Tatsunori Hashimoto, Luke Zettlemoyer, and Mike Lewis. 2022. Contrastive decoding: Open-ended text generation as optimization. *arXiv preprint arXiv:2210.15097*.

Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Xin Zhao, and Ji-Rong Wen. 2023b. Evaluating object hallucination in large vision-language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 292–305, Singapore. Association for Computational Linguistics.

Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252, Dublin, Ireland. Association for Computational Linguistics.

Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9802–9822, Toronto, Canada. Association for Computational Linguistics.

Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. 2023. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9004–9017, Singapore. Association for Computational Linguistics.

Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. 2020. On faithfulness and factuality in abstractive summarization. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 1906–1919, Online. Association for Computational Linguistics.

Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100, Singapore. Association for Computational Linguistics.

Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744.

Vikas Raunak, Arul Menezes, and Marcin Junczys-Dowmunt. 2021. The curious case of hallucinations in neural machine translation. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 1172–1183, Online. Association for Computational Linguistics.

Weijia Shi, Xiaochuang Han, Mike Lewis, Yulia Tsvetkov, Luke Zettlemoyer, and Scott Wen-tau Yih. 2023. Trusting your evidence: Hallucinate less with context-aware decoding. *arXiv preprint arXiv:2305.14739*.

Kurt Shuster, Spencer Poff, Moya Chen, Douwe Kiela, and Jason Weston. 2021. Retrieval augmentation reduces hallucination in conversation. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 3784–3803, Punta Cana, Dominican Republic. Association for Computational Linguistics.

Liam van der Poel, Ryan Cotterell, and Clara Meister. 2022. Mutual information alleviates hallucinations in abstractive summarization. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 3802–3818, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Jonas Waldendorf, Barry Haddow, and Alexandra Birch. 2024. Contrastive decoding reduces hallucinations in large multilingual machine translation models. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2526–2539, St. Julian's, Malta. Association for Computational Linguistics.

Xintong Wang, Jingheng Pan, Liang Ding, and Chris Biemann. 2024. Mitigating hallucinations in large vision-language models with instruction contrastive decoding. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 15840–15853, Bangkok, Thailand. Association for Computational Linguistics.

Yue Zhang, Yafu Li, Leyang Cui, Deng Cai, Lemao Liu, Tingchen Fu, Xinting Huang, Enbo Zhao, Yu Zhang, Yulong Chen, Longyue Wang, Anh Tuan Luu, Wei Bi, Freda Shi, and Shuming Shi. 2023. Siren's song in the AI ocean: A survey on hallucination in large language models. *arXiv preprint arXiv:2309.01219*.